RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS _ C2

Started on	Thursday, 17 September 2026, 8:13 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:18 AM
Time taken	4 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    // Read input from user
    if (scanf("%d", &n) != 1) return 0;

    int count = 0; // The step counter variable
    int c = 0;     // The original algorithm's variable

    // Step Counter Method Placement
    count++; // 1 step for the initialization of the outermost loop (int i = n/2)

    for (int i = n/2; (count++, i < n); i++) {
        for (int j = 1; (count++, j < n); j = 2 * j) {
            for (int k = 1; (count++, k < n); k = k * 2) {
                c++;
                count++; // 1 step for the execution of the body statement (c++)
            }
        }
    }

    // Print the total step count value
    printf("%d\n", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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